another replica instance. Transient errors could also occur on account of network connectivity issues or service unavailability. For applications that are hosted in them, it is imperative that such transient errors are anticipated, and are architected and designed to handle them. The cloud platform would, on its part, provide such indications in the error messages that will state whether that instance of error was transient.

Knowing about the occurrence of such exceptions, system designers should consider certain design patterns that are meant to handle these scenarios in the cloud. For example, a system might leverage 'Retry Pattern' to address a scenario where subsequent requests to a service in the backend might succeed after an initial failed response. Under such circumstances, the client side application should check for that condition and reissue the request. In case the error condition is persistent, the program should back off after a certain number of retries.

7. Always start with a PoC or proof of concept

The dynamics of the cloud are different from those of a local data centre provider. In the case of the latter, an application performs predictably, owing to the years of experience in building, deploying and managing it on premise. Lacunae in the underlying architecture are probably addressed in a custom and ad hoc way. However, any issues faced in the application on premise tend to get amplified when deployed in the cloud.

Hence, when moving to the cloud, however similar the hosting infrastructure is to that in the local data centre, it is a good practice to carry out a 'proof of concept' or PoC to identify any issues or bottlenecks.

8. Understand SLAs

One of the most critical aspects to consider and understand is the SLAs related to individual services in the cloud. While you may offer SLAs to your customers in terms of your solutions HA and DR plan, this cannot be outside of what a public cloud vendor offers. It is illogical to commit to 100 per cent availability to your customers when the cloud vendor is clearly offering SLAs for three 9s (99.9 per cent) or four 9s (99.99 per cent).

As mentioned earlier, transient failures do happen in the public cloud and when that occurs, or when your system goes down, your plan of action is guided by the SLAs you have signed with your clients.

While you build or migrate to the cloud, do take SLAs into account and build your system to handle transient failures and planned/unplanned downtime.

9. Know the environment

Unlike hosting on premise or with a hosting provider where you have taken mere infrastructure/boxes, the public cloud brings in a vibrant compute environment that packages key tenets of a cloud hosting. In addition, with transient failures, it is all the more important to be absolutely clear on troubleshooting tactics, gathering application insights, operating environment statistics and, most importantly, the support model.

You will not be able to design a high-performance and responsive solution in the cloud if you do not realise the host environment's ability on all of the above said attributes. Public cloud providers offer out-of-the-box tools to ensure you stay in close proximity to your deployment and in control.

Awareness is the key here to ensure you are successful in your tryst with the cloud.

10. Simulate and test. 'Make it real'

One of the most challenging aspects of operating in an on-premise environment is the limitation on the amount of compute available to simulate a real environment. The cloud gives you access to unlimited compute due to the humongous investments beings made by cloud providers in setting up data centres globally. Operating in the public cloud environment is hence considered a great opportunity and a choice available for you to go beyond a limited computing boundary in order to test your solution for stability and to identify breaking points.

Living in a world dominated by devices and eventual consumer expectations, your solution needs to be tested for real world load and thresholds. While your business grows, there is no better state to be in than to scale your infrastructure seamlessly. Reality strikes when one realises the limitations of a built solution in a simulated environment, which avoids having to encounter this later, in production.

Simulation is simply about knowing about the breaking points, rather than realising it the hard way when in production. Hence, leverage the cloud for the capacity it offers and 'make it real' by simulating an environment to deal with real world challenges that you may face in the future.

11. Leverage the power of PaaS

The cloud platforms are evolving constantly, and have moved from being mere infrastructure providers to service automation to a great extent—providing backup, recovery and other services necessary to ensure a continuous availability of their applications. With the recent trends in this space, cloud platforms have taken this complexity away from businesses and let them focus only on building their applications, through 'Platform as a Service' (PaaS) offerings.

The notion of PaaS ensures that customers merely focus on building their applications and deploying them, and leave it to the cloud service provider to handle the infrastructure aspects related to hosting and managing scale out/in. In its infancy, PaaS had a disadvantage wherein the applications built for it were locked in with a particular cloud service provider, since the applications had to be compiled for and deployed